State History and Economic Prosperity: New Perspectives from Vietnam

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June 2025

Abstract

The land of what is today Vietnam witnessed the emergence of three early kingdoms during the third century BCE, all were decentralized tributary networks of distinct polities. After resisting the colonization of historical China throughout the first millennium, historical Vietnam started to build a unified nation in the Red River Delta following the Sinic model of political centralization and cultural standardization, while other two kingdoms continued with their decentralized networks. From the 11th to the 18th centuries, historical Vietnam gradually expanded its territory from the Red River Delta to the Mekong River Delta and imposed its model of political centralization and cultural standardization on this new land. Using nighttime light intensity as a proxy for economic prosperity, the present paper uncovers that areas that were governed longer by the centralized state of historical Vietnam are richer in the agricultural economy in the early 1990s, but poorer in the early 2010s after two decades of industrialization. A theoretical framework linking cultural assimilation to economic development can account for this reversal of fortune.

Keywords: State History; Economic Development; Vietnam.

JEL Classification: N45; O43; O53.

[1] I am grateful to Carl-Johan Dalgaard, Claes Ek, Vanessa Kunzmann, Ola Olsson and numerous seminar participants for valuable comments and suggestions. All remaining errors are mine. This research has been funded by the Vietnam National Foundation for Science and Technology Development (NAFOSTED) under grant number 502.01-2024.23.

1 Introduction

Building a nation is an enduring process. The early a society starts the higher its economic prosperity is. In a seminal research, Bockstette, Chanda, and Putterman (2002) show that a longer history of statehood, recorded within the period of 1-1950 CE, is a predictor of modern economic growth and income per capita across countries. Many subsequent cross-country studies, with additional nuances that shall be discussed below, also confirm the empirical regularity between state antiquity and economic prosperity (Chanda and Putterman 2007; Iliev and Putterman 2007; Cinyabuguma and Putterman 2010; Putterman and Weil 2010; Chanda, Cook, and Putterman 2014; Lagerlöf 2016; Borcan, Olsson, and Putterman 2018). The present paper asks a new question: is it just state antiquity that matters, or the types of states also play a role? And use the context of within Vietnam to make an empirical investigation.

During the third century BCE, three kingdoms emerged in the land of what is today Vietnam: historical Vietnam in the Red River Delta, the Champa Kingdom in the central, and the Mekong River Delta was part of historical Cambodia (Figure 1). All three kingdoms shared features typical of Southeast Asian civilization: decentralized tributary networks of independent polities with their own indigenous practices, a balanced economy between wet-rice agriculture and international trade, and a bilateral kinship system. After resisting the colonization of historical China throughout the first millennium CE, historical Vietnam started to build a unified nation following the Sinic model: a centralized bureaucracy, an economy dominated by wet-rice agriculture, and a patriarchal social system. At the same time, the Champa Kingdom and historical Cambodia continued with their Southeast Asian features.

[Figure 1 is about here.]

From the 11th to the 18th centuries, historical Vietnam gradually conquered the Champa Kingdom and then annexed the Mekong River Delta. After establishing military control over an annexed territory, historical Vietnam gradually developed its centralized administrative structure, expanded agricultural production, and forced cultural assimilation. This process of territorial expansion generates a historical experiment in which all areas within today Vietnam share the same length of experiences with state-level institutions, but differ in terms of the duration being governed by the centralized state of historical Vietnam. After a period of French colonization (1858-1945), two Indochina Wars (1945-1975), and a lost decade of the central planning economy, Vietnam emerged in the early 1990s as an underdeveloped agricultural economy. Following a series of market renovations (the Doi Moi), Vietnam has undergone industrialization and registered as one of the fastest growing countries in the world.

Using nighttime light intensity as a proxy for the level of economic development, the present paper uncovers that areas being governed longer by the centralized state of historical Vietnam are richer in the

agricultural economy in the early 1990s, but poorer in the early 2010s after two decades of industrialization. The empirical results also confirm a corollary that these areas experience lower rates of economic growth during this period. Since 1975, all areas within Vietnam have faced the same set of formal institutions, rendering an explanation based on formal institutions impossible. Drawing on Ashraf and Galor (2011), the present paper argues that the centralized state of historical Vietnam promoted cultural assimilation, which was favorable to the development of public infrastructure required for agricultural production. The lack of cultural diversity, however, inhibited the adoption of new knowledge and the transition to industrial production. Exploiting the World Value Survey, the present paper provides tentative evidence that this mechanism can account for the reversal of fortune discovered.

Recent cross-country studies have also uncovered that state antiquity has a non-linear association with economic development. Expanding the history of statehood backwards to 3500 BCE, Borcan, Olsson, and Putterman (2018) find a hump-shaped relationship between state history and economic development. The authors put forward a theoretical framework arguing that, up to a point, longer state history is beneficial for fiscal capacity building, taxation and the provision of public goods, which in turn spur economic growth. But beyond that point, longer state experience is conducive to power centralization, giving rise to extractive institutions that ultimately leads to stagnation.[2] The present paper adds to the literature new perspectives from Vietnam that highlight the importance of state characteristics and the role of cultural diversity in explaining the advent of divergence and overtaking in the process of economic development.

In a related study, Dell, Lane, and Querubin (2018) compare households living on two sides of a discontinued border within the south of Vietnam and find that households on the side being governed longer by the centralized state of historical Vietnam have higher levels of expenditure during the period of 2002-2012. The authors do not examine growth. As a potential mechanism, the authors argue that the centralized state of historical Vietnam enhanced the ability to organize for public goods and redistribution through civil society and local government. By looking at all districts within Vietnam using nighttime light intensity, the present paper is able to capture the full range of values in the history of being governed by the centralized state of historical Vietnam and examine growth. Doing so not only corroborates the finding of Dell, Lane, and Querubin (2018) but also enriches the picture with a newly discovered reversal of fortune.

Studying all districts within Vietnam, Ho, Martinsson, and Olsson (2022) find that areas being governed longer by the centralized state of historical Vietnam are more prone to collectivism with data from 2002 and 2016. Gorodnichenko and Roland (2017) posit that collectivism is favorable to collective actions

[2] See also Lagerlöf (2016) for a similar argument. Across countries outside Europe, Hariri (2012) shows that a longer state history helped a country to resist colonization and prevented European democratic institutions from being transplanted, leaving an autocratic legacy.

while individualism is conducive to innovation, leading to higher levels of technological progress and economic development in individualist societies in the long run. Since collectivism is characterized by prioritizing group norms over individual values (Triandis 1995, Hofstede 2001), the level of collectivism should go hand in hand with the fraction of conformists in a society. Thus, conformism provides a more micro-founded explanation why individualist societies are more favorable to innovation and collectivist societies are better at collective actions.[3] Moreover, the concept of conformism can also account for the advent of divergence and overtaking in the process of economic development (Ashraf and Galor 2011).

The present paper also relates to a well-established literature examining the historical roots of contemporary economic development (Spolaore and Wacziarg 2013; Nunn 2014; Ashraf and Galor 2018), in particular the role of past experiences with statehood (Johnson and Koyama 2017). The next section describes the historical background and lays out a theoretical framework linking cultural assimilation to economic development. Section 3 presents the data and variables in detail. Section 4 reports the empirical results. Section 5 closes the paper with some concluding remarks.

2 Historical Background and Theoretical Framework

2.1 Historical Background

Early States

Archaeological evidence discovers early agricultural communities appearing in the land of what is today Vietnam (Figure 1) during the Neolithic Revolution between 2500 to 1500 BCE (Bellwood 2005, p. 128-145). These communities gradually developed into three separate kingdoms whose written record begins with the arrival of Chinese soldiers and officials in the end of the third century BCE: historical Vietnam in the north, the Champa Kingdom in the central, and the south was part of historical Cambodia (Taylor 1992, p. 137-182). These kingdoms shared many common characteristics of a south-sea civilization spreading across Southeast Asia at that time.

With respect to statecraft, early kingdoms of Southeast Asia were decentralized networks of polities pledging alliances to a powerful center (Hall 2011, p. 1-36). These polities formed tributary relationships with the center, while keeping their own indigenous practices. There were no uniform bureaucratic system and legal codes. Early kingdoms of historical Vietnam centered within the Red River Delta in the north of today Vietnam, while the center of the Champa Kingdom switched between river valley polities along the

[3] A large literature has also pointed out that heterogeneous preferences are generally associated with lower public good provision and greater risk of civil war (Alesina, Baqir, and Easterly 1999; Arbatli et al. 2020; Desmet, Ortuño-Ortín, and Wacziarg 2017).

central coast (Hall 2011, p. 67-102). In the south of today Vietnam, the Mekong Delta region was a peripheral polity of the powerful center of historical Cambodia locating around the Great Lake (Tonle Sap) (Hall 2011, p. 159-210). This region was full of swamps and jungles and was sparsely populated largely by Chams and Khmer (Liere 1980).

Early states of historical Vietnam, the Champa Kingdom, and historical Cambodia maintained a balanced economy between agriculture (primarily wet-rice) and international trade (Hall 2011, p. 67-102, 159-210). Agricultural production relied on the natural ebb and flow of river tides following the monsoon rainy season and local irrigation networks. Within each kingdom, trade involved highland hunters-gatherers exchanging forest products with lowland rice cultivators, in which salt was a key commodity. All kingdoms were active in international trade, initially serving as transitory ports of the trade route between India and China then gradually taking part in the trade as suppliers of local (mostly highland) products (Hall 2011, p. 1-36).

Bilateral kinship was a common feature of early Southeast Asian societies, and was also practiced among early societies of historical Vietnam, the Champa Kingdom, and historical Cambodia (Kirsch 1976, p. 190-202; Hall 2011, p. 1-36; Taylor 1983, p. 1-44). In these societies, inheritance rights could be passed on through both maternal and paternal lines, and women enjoyed high levels of status and freedom relative to men. There is evidence that polygamy was a norm in the ancient society of historical Vietnam, in which there was no stable family system with clearly defined monogamous family units (Taylor 1983, p. 1-44).

Divergence

While the Champa Kingdom and historical Cambodia continued to maintain their Southeast Asian characteristics, historical Vietnam came into contact with the Sinic civilization and was pulled apart. In the first century CE, the Han Empire established its control over historical Vietnam (Taylor 1983, p. 45-84). Throughout the first millennium, different dynasties of historical China had forced their rules into historical Vietnam while facing recurring resistance from the local population (Taylor 1983, p. 85-270). In 939 CE, historical Vietnam gained independence after defeating the Southern Han, and begun the rocky road to building a unified nation. Right at the beginning, historical Vietnam set out to adopt the Sinic model of bureaucratic administration and social organization, at first with a substantial adjustment in order to preserve indigenous practices to gain support from the local population (Taylor 1983, p. 271-301; 1986, p. 139-176). Over time, historical Vietnam increasingly moved closer to the Sinic model of political centralization and cultural standardization (Lieberman 2003, p. 338-456; Taylor 2013, p. 51-445; Whitmore 1997, p. 665-687).

To grow a national identity, historical Vietnam developed a centralized bureaucratic system with a uniform administrative structure down to the village level and legal codes across the country (Lieberman 2003, p. 338-456; Taylor 2013, p. 51-445). This process of integrating local communities into the national space was not linear, experienced recurring periods of slowdowns following the waxing and waning of dynastic power (Nguyen 2003; Yu 2001).[4] Taxation was also uniform across the country, consisting mainly of agricultural land taxes and head taxes, together with other secondary taxes and duties (Tana 1998, p. 48-56). A national civil service examination, open to all and increasingly drawing on Confucianism over time, was organized to recruit government officials (Whitmore 1968, p. 114-130; Woodside 1988, p. 169-233).

Early dynasties of historical Vietnam put more emphasis on expanding and developing their agricultural production in the Read River plain relative to international trade (Hall 2011, p. 89-102). Private ownership was granted to newly cleared land in order to create incentives for peasants to develop new land for agricultural production (Ho 2023). Unpaid labor duties were imposed on adult males to construct a centralized irrigation system supporting agricultural expansion, a national road network connecting all regions to the central court, as well as other public infrastructure (Tana 1998, p. 48-56). Over time, foreign trade in general was decreasing and tax revenue collected from imports and exports became negligible (Woodside 1988, p. 261-276).

At the beginning of its independence, historical Vietnam preserved the indigenous practices of polygamy and bilateral kinship (Taylor 1983, p. 271-301; 1986, p. 139-176). Over time, successive dynasties gradually brought the society closer to a patriarchal social system with clearly defined monogamous family units (Whitmore 1984). A systematic and comprehensive set of rules and codes of conduct were developed and promoted through education and civil service examination to regulate almost all aspects of life, including relationships between the king and his people, husband and wife, parents and child, among many others (Yu 1990, p. 81-231; Woodside 1988, p. 7-111). This cultural standardization was crucial to developing and maintaining public infrastructure, as well as enhancing social stability (Whitmore 1984, 1997).

The Vietnamese Southern Advance

Following the process of political centralization and cultural standardization, historical Vietnam witnessed agricultural expansion and population growth, which in turn helped it defending against powerful invaders such as the Mongols. From the 11th to the 18th centuries, historical Vietnam gradually conquered the Champa Kingdom and annexed the Mekong River Delta (Figure 1). This so-called southern advance (*nam tien*)

[4] This gives rise to the cliché: “The laws of the king are second to the customs of the village”.

happened in many steps and completed in 1757, by which historical Vietnam already controlled a territory as large as today.[5] In each step, historical Vietnam first conquered and imposed military control over an annexed territory, then gradually established its centralized administrative structure, expanded agricultural production, and forced cultural assimilation (Whitmore 2014, p. 120-143; Wook 2003, p. 47-65). To strengthen the political and social integration, historical Vietnam also encouraged its citizens to migrate and settle in newly annexed territories (Tana 1998, p. 18-36).

The modern period of Vietnamese history begins with the French colonization, which started in 1858 and ended with the Vietnamese victory in the First Indochina War (1946-1954). Following the French defeat was the American intervention in Vietnam, i.e., the Second Indochina War (commonly known as the Vietnam War), which was ended by the reunification of the country in 1975. After that, the central planning economy was implemented in the whole country with limited success until the end of 1980s. Vietnam appeared in the early 1990s as an underdeveloped agricultural economy and started to implement a series of market renovations (the Doi Moi). Since then, Vietnam has undergone industrialization and registered as one of the fastest growing countries in the world.

2.2 Theoretical Framework

The forces of political centralization and cultural assimilation spreading throughout historical Vietnam along with the process of territorial expansion and integration could have differential influences on contemporary economic development across regions. Although all areas within today Vietnam experienced approximately the same length of state history, they differ substantially in terms of the length being governed by the centralized state of historical Vietnam. As discussed above, being exposed longer by forces of political centralization and cultural assimilation should lead to a higher level of cultural homogeneity, which in turn might influence economic development.

The present paper borrows the seminal work of Ashraf and Galor (2011) to sketch out a theoretical framework, with testable hypotheses, linking cultural homogeneity to economic development in the context of Vietnam. To start with, there are conformists and non-conformists with respect to norms in a society. There are also forces of cultural assimilation that increase the fraction of conformists, and forces of cultural diffusion that expand the fraction of non-conformists. Cultural assimilation and cultural diffusion play opposite roles in production. On the one hand, non-conformists are conducive to the generation and adoption of new knowledge in the economy, which in turn promotes technological progress. On the other hand, non-conformists obstruct the transmission of knowledge specific to a society within and across

[5] See Ho, Martinsson, and Olsson (2022) for a detailed chronicle of this southward territorial expansion.

generations, which in turn reduces its ability to exploit the existing technological frontier. This knowledge includes technological know-how and social norms, among others.

In early stages of economic development, the level of productivity of the manufacturing sector is so low that it was not economically viable, thus only the agricultural sector is operative. In this agricultural economy, an increase in the fraction of conformists has a beneficial effect on the level of productivity in the agricultural sector, because its positive influence on the transmission of society-specific knowledge outweighs its negative role in the advancement of agricultural productivity. Owing to stronger forces of cultural assimilation, areas being governed longer by the centralized state of historical Vietnam enjoyed a higher fraction of conformists and a greater efficiency in the transmission of society-specific knowledge. This knowledge included agricultural know-how and social norms conducive to cooperation and collective actions in developing and maintaining agricultural infrastructure. As a result, these areas were better at exploiting the agricultural technological frontier and had a higher level of economic development. This gives rise to the first hypothesis: *in the agricultural economy, areas being governed longer by the centralized state of historical Vietnam had higher levels of economic development*.

When the productivity of the latent manufacturing sector is high enough, a society with a larger but still moderate fraction of non-conformists witnesses the beneficial effect in the advancement of industrial productivity, relative to that in agricultural, outweighs its eroding effect on the transmission of society-specific knowledge. In contrast, the associated lack of cultural diffusion delayed the advancement of knowledge in a conformist society, delaying it from shifting to an industrial technology. Owing to stronger forces of cultural diffusion, areas being governed shorter by the centralized state of historical Vietnam enjoyed a higher fraction of non-conformists and a greater adoption of new knowledge. This knowledge included industrial technologies, market principles, and foreign practices. As a result, the areas witnessed increases in productivity and catch-up growth, finally surpassed areas being governed longer by the centralized state of historical Vietnam in terms of level of economic development. This leads to the second hypothesis: *after the period of industrialization, areas being governed longer by the centralized state of historical Vietnam had lower levels of economic development*. The third hypothesis appears as a corollary: *during decades of industrialization, areas being governed longer by the centralized state of historical Vietnam experienced lower rates of economic growth*.

3 Data

3.1 Economic Development

To investigate three hypotheses discussed above, the ideal (but impossible) approach is to measure gross domestic product (GDP) at district level going back to 1850s, the time before Vietnam was colonized by France. The second-best strategy is to use GDP in the early 1990s and examine how experiences with the French colonization and subsequent events affect the empirical results. Unfortunately, GDP is not measured at the district level in Vietnam. Another approach to measure economic development is to use household expenditure taken from the Vietnam Household Living Standards Survey (VHLSS). This survey, however, was not sampled to represent information at the district level. Moreover, it was only collected for a large sample of districts starting from 2002, a decade after Vietnam started the market reform and the process of industrialization in the early 1990s.

Drawing on a recent innovation, the present paper uses nighttime light intensity from the Defense Meteorological Satellite Program (DMSP) database to capture economic development at the district level.[6] The database is provided by the National Oceanic and Atmospheric Administration (NOAA) for the period of 1992-2013. NOAA processes satellite images to calculate annual nighttime light intensity produced by man-made light, i.e., removing observations affected by cloud coverage or polar lights and setting observations that are likely to reflect fires, other ephemeral lights, or background noise to zero (Baugh et al. 2010). The nighttime light intensity value is increasing on a scale from 0 to 63.[7] When light intensity reaches higher than 63, it is top-coded. Nighttime light intensity at the district level is calculated by taking the average of the values of all cells belonging to each district. The present paper use data for the beginning (1992) and ending (2013) year of the series.

[Table 1 is about here.]

Figure 2 plots nighttime light intensity in 1992 and 2013 for 627 districts with available data (out of 630 districts in total). The first impression is that Vietnam in general was a dark country at night in 1992 (Panel A), with an average nighttime light intensity of 3.41 and the maximum value of 62.42 (Table 1). Also in 1992, areas surrounding the Red River Delta in the north were in general brighter than areas surrounding the Mekong River Delta in the south. With sustained economic growth, Vietnam was much brighter in 2013 (Panel B), with an average nighttime light intensity of 12.06 and the maximum value of 63 (Table 1). Figure 2 shows a visual reversal of fortune, in which areas surrounding the Mekong River Delta in the south

[6] Many studies have shown that nighttime light intensity is a reasonable proxy for economic development (Henderson, Storeygard, and Weil 2012; Hodler and Raschky 2014; Michalopoulos and Papaioannou 2013, 2014; Nordhaus and Chen 2015).

[7] In empirical studies, nighttime light intensity is normally transformed into logarithmic scale to minimize the problem of outliers, with zero-value observations retained by adding 0.01. The argument is that zero values does not reflect no nighttime light at all, and certainly does not imply an absence of economic activities given that all administrative areas are populated. They are an artificial product of data collection and processing. In particular, there are certainly man-made nighttime lights in districts with zero values, but they might be too low to be detected by the satellites.

became brighter in general than areas surrounding the Red River Delta in the north. Whether or not this visual finding survives more rigorous empirical analysis shall be answered in the following section.

[Figure 2 is about here.]

3.2 Centralized State Antiquity and Control Variables

Centralized State Antiquity

The main explanatory variable is the time elapsed since a district was governed by the centralized state of historical Vietnam, which is taken from Ho, Martinsson, and Olsson (2022). The authors code an area as being governed by the centralized state of historical Vietnam when there is an official record saying that this area was annexed and under control of historical Vietnam. Historical areas are then matched with modern districts based on various studies on historical Vietnamese border.[8] Following the earlier historical discussion, the onset of the first unified state of historical Vietnam in 968 is taken as the year that political centralization and cultural standardization started to have impacts on the people of historical Vietnam. The ending year is set at 1990. In the current sample of 627 districts, the time since being governed by the centralized state of historical Vietnam is averaged at 6.82 centuries (Table 1).

Control Variables

There are potential confounding factors that might influence both the time since being governed by the centralized state of historical Vietnam and the level of economic development. First, it is argued that agricultural suitability was an important motive behind territorial expansions of empires such as historical China and historical Vietnam (Hall 2011, p. 67-102; Taylor 1983, p. 1-44). At the same time, regions suitable for agriculture and earlier Neolithic transitions initially experienced higher population densities but eventually lagged in industrialization owing to limited technological spillovers and low demand for human capital in agriculture (Ashraf and Galor 2013; Galor 2011, 2022; Galor and Mountford 2008).[9] In a similar manner, other important motives might include irrigation suitability and climatic conditions, which were crucial for agricultural production. Second, geographical isolation might make an area harder to conquer. It

[8] As an example, the Champa Kingdom submitted its land in O and Ly as wedding presents to historical Vietnam during a marriage in 1306. Based on Dao (2005, p. 163-164), O was in the south of Thach Han River in modern Quang Tri province and Ly was modern Thua Thien Hue province. Thus, all districts in the south of Thach Han River in modern Quang Tri province (Trieu Phong, Quang Tri, Da Krong, and Hai Lang) and in modern Thua Thien Hue province are coded as being governed by the centralized state of historical Vietnam since 1306.

[9] The present paper only considers agricultural suitability because data on the timing of the Neolithic transition are unavailable at subnational levels in Vietnam. Nevertheless, archaeological evidence shows that early agricultural communities appeared across modern Vietnam around roughly the same time (Bellwood 2005, p. 128-145).

is also conducive to cultural assimilation (Ashraf and Galor 2011) and influence economic development (Nunn and Puga 2012). To capture geographical isolation, the present paper uses distance to the coast, elevation, and ruggedness.

Agricultural suitability is measured by the caloric suitability index constructed at 5 arcminute resolution by Galor and Özak (2016). The authors base their calculation on data from the Global Agro-Ecological Zones project of the Food and Agriculture Organization. The caloric suitability index measures the average potential yield (million kilo calories per squared kilometer per year) attainable in each grid cell given the set of crops that are suitable for cultivation in the post-1500 period. To capture the natural component of land productivity, i.e., unaffected by human intervention, the production conditions are set at a low level of inputs and rain-fed agriculture based on agro-climatic conditions.

The Food and Agriculture Organization defines irrigation suitability as the potential increase in agricultural output that can be obtained by fully exploiting irrigation compared to rain-fed agriculture, and classifies irrigation suitability into five classes: (1) only suitable for rain-fed agriculture, (2) output yield increases by 0-20%, (3) output increases by 20-50%, (4) output increases by 50-100%, and (5) output increases by more than 100% (Fischer et al. 2002). Following Buggle (2020), we classify an area to be suitable for irrigation if agricultural output increases by at least 50% to construct an irrigation suitability index. Climatic zones are defined by the Köppen-Geiger classification, and are constructed using precipitation and temperature data in the period of 1901-1925 (Rubel and Kottek 2010).

Distance to the coast is measured by the shortest (bird-fly) distance to the coastal line. Elevation is taken from the Global 30 Arc-Second Elevation Dataset (GTOPO30) provided by the Earth Resources Observation and Science Center. The terrain ruggedness index was originally devised by Riley, DeGloria, and Elliot (1999) to quantify topographic heterogeneity in wildlife habitats providing concealment for prey and lookout posts. This index is calculated by Nunn and Puga (2012) based on the GTOPO30 dataset. Table 1 reports descriptive statistics of all variables.

Table A1 in the Appendix shows that all potential confounding factors are significantly correlated with the time since being governed by the centralized state of historical Vietnam. Distance to the coast, elevation, and ruggedness have positive Pearson's correlation coefficients, whereas agricultural suitability and irrigation suitability have negative ones. As discussed in the historical background, historical Vietnam gradually expanded its territory from the Red River Delta to the Mekong River Delta following a north-south geographical order. Thus, the time since being governed by the centralized state of historical Vietnam is positively correlated with latitude. This correlation poses a challenge to the empirical analysis, which will be examined in detail later in the robustness check.

4 Empirical Analysis

4.1 Baseline Results

To examine each hypothesis discussed above, I estimate the following regression model:

$$Eonomic\ Development_d = \alpha + \beta Centralized\ State\ Antiquity_d + \gamma' \mathbf{X}_d + \epsilon_d, \qquad (1)$$

in which *Economic Development*$_d$ is the corresponding measure of economic development in district d, *Centralized State Antiquity*$_d$ is the number of centuries being governed by the centralized state of historical Vietnam, $\mathbf{X}_d$ is a set of potential confounding factors, and ϵ_d is a random error term.

To test the first hypothesis that in the early 1990s agricultural economy, areas being governed longer by the centralized state of historical Vietnam had higher levels of economic development, I use the natural logarithm of nighttime light intensity in 1992 as the dependent variable. Table 2 shows that the estimated coefficients of centralized state antiquity are positive and significant without and with the inclusion of potential confounding factors (Columns 1 and 8). On average, districts that are governed longer by the centralized state of historical Vietnam have higher level of nighttime light intensity in 1992. The marginal effect is large, in the full specification (Column 8), one century increase in being governed by the centralized state of historical Vietnam is associated with roughly 11.4% higher in nighttime light intensity in 1992.

[Table 2 is about here]

Turning to the second hypothesis that after two decades of industrialization, areas being governed longer by the centralized state of historical Vietnam had lower levels of economic development, I use the natural logarithm of nighttime light intensity in 2013 as the dependent variable. Table 3 reports that the estimated coefficients of centralized state antiquity are negative and significant without and with the inclusion of potential confounding factors (Columns 1 and 8). On average, districts that are governed longer by the centralized state of historical Vietnam have lower level of nighttime light intensity in 2013. The marginal effect is large, in the full specification (Column 8), one century increase in being governed by the centralized state of historical Vietnam is associated with roughly 9% lower in nighttime light intensity in 2013.

[Table 3 is about here]

To examine the third hypothesis that during two decades of industrialization, areas being governed longer by the centralized state of historical Vietnam experienced lower rates of economic growth, I use the growth rate of nighttime light intensity in the period of 1992-2013 as the dependent variable. This growth rate is approximated by taking the difference between the natural logarithm of nighttime light intensity in 2013 and the natural logarithm of nighttime light intensity in 1992. To account for the effect of diminishing

marginal return, i.e., richer districts tend to grow slower (Solow 1956), I add the natural logarithm of nighttime light intensity in 1992 to the set of potential confounding factors. Table 4 shows that the estimated coefficients of centralized state antiquity are negative and significant without and with the inclusion of potential confounding factors (Columns 1 and 8). On average, districts that are governed longer by the centralized state of historical Vietnam have lower growth rate of nighttime light intensity during the period of 1992-2013. The marginal effect is large, in the full specification (Column 8), one century increase in being governed by the centralized state of historical Vietnam is associated with roughly 14.2% lower in the growth rate of nighttime light intensity during the period of 1992-2013. In addition, the estimated coefficients of the natural logarithm of nighttime light intensity in 1992 are negative and significant (Columns 1 and 8), confirming the effect of diminishing marginal return.

[Table 4 is about here]

With respect to the potential confounding factors, the estimated coefficients of the caloric suitability, irrigation suitability, climatic zones and elevation do not have consistent signs and statistical significance across specifications in all three hypotheses (Table 1, 2, and 3). Only the estimated coefficients of distance to the coast and ruggedness are consistently negative and significant. On average, districts locating father from the coast and having more rugged terrains have lower level of nighttime light intensity in 1992 and 2013, as well as lower growth rate of nighttime light intensity during the period of 1992-2013. This finding indicates a geographical persistence of fortune.

4.2 Robustness

Top-coding

As discussed above, nighttime light intensity is top-coded at 63, failing to reflect the true brightness of highly-developed cities. Bluhm and Krause (2022) shows theoretically and empirically that lights in these cities can be characterized by a heavy-tailed Pareto distribution, which can be used to recover the full distribution of lights. As expected in the context of Vietnam, correcting for top-coding has a substantial impact on average nighttime light intensity in 2013 (Table 1). To examine the robustness of the baseline results with respect to top-coding, I re-estimate all specifications using this corrected series of nighttime light intensity. Table 5 (Panel A) shows that the estimated coefficients of centralized state antiquity remain qualitatively unchanged. As expected, the marginal effects are similar to the baseline results with respect to nighttime light intensity in 1992, but larger with respect to nighttime light intensity in 2013.

[Table 5 is about here]

Lowland Areas

It is well-known that historical Vietnam was primarily interested in governing the lowland areas that were suitable for wet-rice agriculture, and hence only imposed tributary relationships with people from highland areas (Lieberman 2003, p. 338-456; Taylor 2013, p. 51-445). To examine whether or not this differential treatment between lowland and highland areas drives the baseline results, I exclude from the sample all districts in the highland areas, i.e., districts whose average elevations are above 500 meters.[10] Table 5 (Panel B) reports that the estimated coefficients of centralized state antiquity remain qualitatively unchanged.

Omitted Variables

How far are the baseline results are driven by omitted variables? One way to examine this problem is to use the change in the estimated coefficient of centralized state antiquity when observed confounding factors are included to infer what would happen if unobserved confounding factors were to be added. Oster (2019) develops a consistent estimate adjusted for omitted-variable bias under the assumption that the selection on unobserved confounding factors is proportional to the selection on observed confounding factors. Let the selection on unobserved confounding factors be one half of the selection on observed confounding factors, the adjusted estimated coefficients of centralized state antiquity remain sizable in magnitude compared to their unadjusted counterparts (Columns 8 of Tables 2, 3, and 4).

Latitude

The latitude gradient appears to be the most mysterious empirical regularity in cross-country comparative development, i.e., contemporary economic prosperity on average increases as one moves away from the equator. Conditioning on other geographical variables, Andersen, Dalgaard, and Selaya (2016) show that latitude has a strong association with the intensity of ultraviolet radiation (UR-V). UR-V causes debilitating eye diseases, reducing returns to skill investments and lowering contemporary economic development. Because within Vietnam latitude only ranges from 23.24 in the northernmost to 8.65 in the southernmost districts, I argue that this channel is negligible in explaining economic prosperity across districts. Moreover, UR-V cannot theoretically account for the reversal of fortune found above with respect to the time since being governed by the centralized state of historical Vietnam.

As discussed in the historical background, historical Vietnam had to expand its territory following a north-south geographical order because it could not conquer the Mekong River Delta in the south without

[10] The results are robust to other values such as 400 and 600 meters.

annexing all areas located in between.[11] Therefore, latitude can also affect economic prosperity across districts within Vietnam through its impact on the time since being governed by the centralized state of historical Vietnam. If this is the case then latitude should be able to directly account for the reversal of fortune found above. To examine the channels through which latitude affects economic prosperity across districts within Vietnam, I first use latitude as an explanatory variable. The estimated results in Table 6 (Panel A) indicates that districts with higher latitudes on average have higher level of nighttime light intensity in 1992, lower level of nighttime light intensity in 2013, and lower growth rate of nighttime light intensity during the period of 1992-2013 (Columns 1, 3, and 5). When latitude and the time since being governed by the centralized state of historical Vietnam are included together, the estimated coefficients of both variables change signs and lose significance. These findings indicate that latitude affect economic prosperity across districts within Vietnam mainly through its impact on the time since being governed by the centralized state of historical Vietnam.

[Table 6 is about here]

If one is willing to walk a little farther with the assumption that latitude only affects economic development through its effect on centralized state antiquity, then it would be useful to use the north-south geographical order captured by latitude as an instrumental variable for centralized state antiquity. This exclusion restriction assumption enables another robustness check with respect to omitted-variable bias. Table 6 (Panel B) shows that the estimated coefficients of centralized state antiquity remain qualitatively unchanged and similar in magnitude compared to the baseline results.[12]

Chinese colonization

One may argue that the forces of political centralization and cultural assimilation in historical Vietnam had arrived with the Chinese colonization, almost a millennium before the first independent centralized state of historical Vietnam was established. This colonial centralization and assimilation might have served the aims of subjugation and extraction, resulting in different economic trajectories in the long run. To examine this issue, I add the duration being ruled by the Chinese colonization to the time since being governed by the centralized state of historical Vietnam for those districts that fell under the rule by this colonization.[13] Panel

[11] In theory, historical Vietnam could conquer the Mekong River Delta by marine route along the coastal line in the east or mountainous route in the west. Both strategies were infeasible given the historic technologies of logistics and transportation.

[12] The first-stage results ensure that latitude is a strong predictor of centralized state antiquity, i.e., significantly positive estimated coefficients of latitude and large *F* statistics (the whole results can be produced with the replication package).

[13] Taylor (1983, p. 45-84) shows that the Han Empire started building up its administration in the Red River Delta since 43 CE.

A of Table A2 in the Appendix shows that the estimated coefficients of the adjusted centralized state antiquity remain qualitatively unchanged. This finding indicates that the Chinese colonization contributed to the political centralization and cultural assimilation in ways that complement later independent states of historical Vietnam.

Spatial Dependence

To address concern with respect to spatial dependence, Panel B of Table A2 in the Appendix reports results using standard errors adjusted for spatial autocorrelation following Conley (1999). The assumption is that autocorrelation decreases in the distance between district centroids and equals zero for districts that are more than 0.25 degree (roughly 25 kilometers) apart. A distance of 25 kilometers connects two averaged districts in Vietnam. As can be seen, the estimated coefficients of centralized state antiquity remain unchanged with respect to statistical significance.

4.3 Mechanism

The empirical results so far have lent support to three hypotheses discussed earlier. The forces of political centralization and cultural assimilation of the centralized state of historical Vietnam provide advantages in an agricultural economy in the 1990s. Only during the next two decades of rapid industrialization, the table turns as cultural homogeneity prevents a quick adoption of new knowledge and practices, resulting in a reversal of fortune. This section explores cultural assimilation as well as other complementary mechanisms behind these results.

Cultural diversity

As discussed in the theoretical framework, cultural diversity, expressed in the fraction of conformists, plays a central role in explaining the reversal of fortune with respect to economic development. The forces of political centralization and cultural assimilation of the centralized state of historical Vietnam reduced cultural diversity, or increased the prevalence of conformists. One critical aspect of this loss of cultural diversity is the subjugation of indigenous practices of bilateral kinship by a patriarchal social system. Another aspect is the replacement of the practice of polygamy with no stable family system with clearly defined monogamous family units, giving the family a central role in the social organization. Thus, there should be positive associations between the time since being governed by the centralized state of historical

Vietnam and the prevalence of conformists with respect to men-dominated gender norms and the importance of the family.

The World Value Survey in Vietnam, with location information of respondents recorded at the province level in the 2006 round, asks a question about the role of the women: "*Being a housewife is just as fulfilling as working for pay*". I code the answers into *agree* (1) and *disagree* (0) to capture conformists.[14] There is another question about the importance of the family: "*How important is the family in your life?*" I code the answer into *important* (1) and *not important* (0) to capture conformists.[15] The sample consists of around 1500 respondents from 20 (out of a total of 64) provinces. Roughly 77% of respondents agree to the statement about the role of the women, and around 81% of respondents regard family as important.

To examine the relationship between centralized state antiquity and the prevalence of conformists, I estimate the following regression models:

$$Conformist_{ip} = \alpha + \beta Centralized\ State\ Antiquity_p + \gamma' \mathbf{X}_p + \epsilon_{ip}, \qquad (2)$$

in which $Conformist_{ip}$ indicates individual i living in province p is a conformist, $Centralized\ State\ Antiquity_p$ is the number of centuries that province p being governed by the centralized state of historical Vietnam, $\mathbf{X}_p$ is a set of potential confounding factors, and ϵ_{ip} is a random error term. Table 7 reports that the estimated coefficients of centralized state antiquity are positive and significant without and with the inclusion of potential confounding factors. On average, respondents living in provinces that are governed longer by the centralized state of historical Vietnam are more likely to be conformists with respect to men-dominated gender norms and the importance of the family. The marginal effects are sizable, i.e., one century increase in the time being governed by the centralized state of historical Vietnam is associated with roughly 2% higher in the likelihood of being a conformist with respect to gender norms, and around 3.5% higher with respect to the importance of the family.

[Table 7 is about here]

Although the empirical results presented above lend support to cultural diversity as an explanation for the economic development and reversal of fortune within Vietnam, a legitimate concern points out that gender norms and the importance of the family are themselves influenced by economic development. While there is a certain amount of coevolution between these variables, I argue that a significant part of causation should go from cultural diversity to economic prosperity. This is because differences in cultural diversity across historical Vietnam had already existed even before the modern economy. For example, studying

[14] *Agree* = *Strongly agree* or *Agree*, *Disagree* = *Disagree* or *Strongly disagree*.
[15] *Important* = *Very important*, *Not important* = *Rather important* or *Not very important*. No respondents chose "*Not at all important*".

Vietnam in the eighteenth century, Tana (1998, p. 99-118) provides ample qualitative evidence that men-dominated gender norms and the importance of the family were stronger in areas being governed longer by the centralized state of historical Vietnam.

Western contacts

When French colonizers arrived in Vietnam, they imposed the south under their direct rule as the colony of Cochinchina. Whereas the north and the central of Vietnam became French protectorates in which the king of Vietnam retained his rule under French supervision. After the First Indochina War (1946-1954) and the retreat of French colonizers, Cochinchina became the base of American occupation during the Second Indochina War (1955-1975). These Western contacts might have changed the south of Vietnam in ways that shape modern economic prosperity. One can also argue that French colonizers chose the south of Vietnam because this region, being ruled by the centralized states of historical Vietnam later, featured a higher level of cultural diversity relative to other regions. Thus, these Western contacts can be another channel through which the time since being governed by the centralized state of historical Vietnam affect modern economic prosperity.

To examine the influence of the French colonization and American occupation, I include a dummy for Cochinchina in the empirical analysis. Panel A of Table A3 in the Appendix shows that the estimated coefficients of Cochinchina are positive and significant when entering the regressions alone (Columns 1, 3, and 5), but turn negative when all other variables are included. Meanwhile, the estimated coefficients of centralized state antiquity remain qualitatively unchanged (Columns 2, 4, and 6). This finding suggests that the French Colonization and the two Indochina Wars did not erase the shadow of the centralized states of historical Vietnam to a large extent.

Post-war investments

Another potential explanation for the reversal of fortune found after 1990s is public investments, i.e., the government might have made relatively higher levels of investments in districts that have lower levels of the time since being governed by the centralized state of historical Vietnam. These differential investments might have generated higher growth rates and eventually led to the reversal of fortune. Nevertheless, data on public investments are only available at the province level since 2010. To examine the role of public investments, I collect these data for all provinces that report them in their statistical yearbooks in 2010, 2011, and 2012. Panel B of Table A3 in the Appendix shows that the estimated coefficients of centralized state antiquity are insignificant in explaining public investments per capita at the province level for all three

years. This finding suggests that post-war public investments do not play an important role in explaining the reversal of fortune found after 1990s.

State capacity

Even though formal institutions cannot account for sub-national differences in economic performance within Vietnam, state capacity of the local government, defined as the ability to enforce rules and carry out policies, may have a role to play. In the spirit of Borcan, Olsson, and Putterman (2018), areas being governed longer by the centralized state of historical Vietnam might have relatively stronger state capacity in the 1990s, which in turn were conducive to economic development. Then during the next two decades, state capacity of these areas might deteriorate, resulting in relatively lower rates of economic growth and an overtaking by areas being governed shorter by the centralized state of historical Vietnam. If this is the case, then areas being governed longer by the centralized state of historical Vietnam should have relatively weaker state capacity in the early 2010s.

To examine this mechanism, state capacity can be captured by the Vietnam Provincial Governance and Public Administration Performance Index (PAPI). PAPI is constructed and reported by the United Nations Development Program (UNDP) since 2011 to measure experiences and perceptions of citizens about the quality of policy implementation and service delivery of all 63 provincial governments in Vietnam.[16] Panel C of Table A3 in the Appendix shows that the estimated coefficients of centralized state antiquity are positive, indicating that provinces being governed longer by the centralized state of historical Vietnam on average have stronger state capacity in the early 2010s. Nevertheless, these coefficients are not significant in all specifications. Taken together, these findings suggest that state capacity does not play an important role in explaining the divergence and overtaking within Vietnam found after the 1990s.

5 Conclusion

Numerous cross-country studies have found a positive association between state history (since 1 CE) and economic prosperity. Recent studies have expanded the literature by discovering a hump-shaped relationship between the history of statehood (back to 3500 BCE) and the level of economic development, i.e., a young state often lacks administrative capacity and public goods while an old state often falls prey to excessive centralization and extractive institutions. The present paper examines if the types of state also

[16] It consists of six dimensions: (1) civic participation at local levels, (2) transparency of local decision-making, (3) vertical accountability, (4) control of corruption, (5) public administrative procedures, and (6) public service delivery. Principal component analysis is used to construct an overall (standardized) index.

matter for economic development by exploiting a historical experiment within Vietnam that generated variations in the duration that a modern area being governed by the centralized state of historical Vietnam.

Using nighttime light intensity as a proxy for the level of economic development, the present paper uncovers that areas that were governed longer by the centralized state of historical Vietnam are richer in the agricultural economy in the early 1990s, but poorer in the early 2010s after two decades of industrialization. Suggestive evidence also indicates that the forces of political centralization and cultural standardization reduced cultural diversity, which in turn driving this reversal of fortune. These findings add to the literature new perspectives from Vietnam that highlight the importance of state characteristics and the role of cultural diversity in explaining the advent of divergence and overtaking in the process of economic development.

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Table 1. Variable Description

Variable	Description	Mean	SD	Min.	Max.	N
Main variables						
Light intensity 1992	Average nighttime light intensity in 1992	3.41	9.81	0	62.42	627
Light intensity 2013	Average nighttime light intensity in 2013	12.06	16.30	0	63	627
Light intensity 1992 (top)	Average nighttime light intensity in 1992 (corrected for top-coding)	4.71	20.93	0	284.80	627
Light intensity 2013 (top)	Average nighttime light intensity in 2013 (corrected for top-coding)	20.89	64.40	0	825.20	627
Centralized state antiquity	Centuries governed by the centralized state of historical Vietnam	6.82	3.56	2.33	10.22	627
Control variables						
Caloric suitability	Average yield (million kilo calories per km^2 per year)	25.08	2.49	15.88	31.74	627
Irrigation suitability	Average suitability for irrigation, not suitable (0) and suitable (1)	0.09	0.24	0	1	627
Am climatic zone	Equatorial monsoonal	0.19	0.39	0	1	627
Aw climatic zone	Equatorial winter dry	0.46	0.50	0	1	627
Cfa climatic zone	Warm temperature, fully humid, hot summer	0.01	0.12	0	1	627
Cwa climatic zone	Warm temperature, winter dry, hot summer	0.33	0.47	0	1	627
Distance to coast	Shortest distance to the coastal line (100 km)	0.74	0.82	0	4.33	627
Elevation	Average height above sea level (km)	0.21	0.31	0.001	1.56	627
Ruggedness	Average topographic heterogeneity (10000 km)	1.20	1.46	0	6.27	627
Latitude	Latitude	16.50	4.78	8.65	23.24	627

Note: All variables are measured at the district level. See the main text for more detail.

Table 2. Economic Development in the Agricultural Regime

	Ln Nighttime Light Intensity 1992							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centralized state antiquity	0.104***							0.114***
	(0.030)							(0.035)
Caloric suitability		-0.017						-0.192***
		(0.038)						(0.047)
Irrigation suitability			0.394					-0.274
			(0.454)					(0.480)
Aw climatic zone				1.377***				0.152
				(0.245)				(0.226)
Cfa climatic zone				-0.881				0.175
				(0.607)				(0.525)
Cwa climatic zone				1.495***				1.218***
				(0.283)				(0.269)
Distance to coast					-0.969***			-0.421**
					(0.099)			(0.169)
Elevation						-4.558***		-0.052
						(0.295)		(0.630)
Ruggedness							-1.028***	-1.173***
							(0.055)	(0.120)
Centralized state antiquity (adj.)								0.120
R^2	0.019	0.000	0.001	0.052	0.087	0.269	0.311	0.441
Observations	627	627	627	627	627	627	627	627

Note: Robust standard errors are in parentheses. All regressions include a constant. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. The base climatic zone is Am. Centralized state antiquity (adj.) represents the estimated coefficient of Centralized state antiquity adjusted for omitted-variable bias following the method developed by Oster (2019)
* p<0.1, ** p<0.05, *** p<0.001

Table 3. Economic Development in the Age of Industrialization

	Ln Nighttime Light Intensity 2013							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centralized state antiquity	-0.127***							-0.089***
	(0.022)							(0.023)
Caloric suitability		0.150***						-0.203***
		(0.038)						(0.046)
Irrigation suitability			1.141***					-0.636***
			(0.246)					(0.239)
Aw climatic zone				0.734***				-0.377***
				(0.177)				(0.135)
Cfa climatic zone				-0.782				0.227
				(0.747)				(0.520)
Cwa climatic zone				-0.408*				0.268
				(0.242)				(0.176)
Distance to coast					-1.367***			-0.534***
					(0.106)			(0.168)
Elevation						-4.576***		-0.231
						(0.321)		(0.514)
Ruggedness							-1.085***	-1.011***
							(0.048)	(0.094)
Centralized state antiquity (adj.)								-0.071
R^2	0.047	0.032	0.018	0.063	0.288	0.449	0.573	0.614
Observations	627	627	627	627	627	627	627	627

Note: Robust standard errors are in parentheses. All regressions include a constant. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. The base climatic zone is Am. Centralized state antiquity (adj.) represents the estimated coefficient of Centralized state antiquity adjusted for omitted-variable bias following the method developed by Oster (2019)
* p<0.1, ** p<0.05, *** p<0.001

Table 4. Growth during Industrialization

	Nighttime Light Intensity Growth 1992-2013							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centralized state antiquity	-0.193***							-0.142***
	(0.013)							(0.017)
Ln light intensity 1992	-0.357***							-0.537***
	(0.019)							(0.019)
Caloric suitability		0.167***						-0.114***
		(0.023)						(0.035)
Irrigation suitability			0.747***					-0.509***
			(0.269)					(0.120)
Aw climatic zone				-0.643***				-0.447***
				(0.160)				(0.102)
Cfa climatic zone				0.099				0.146
				(0.580)				(0.422)
Cwa climatic zone				-1.903***				-0.296**
				(0.160)				(0.141)
Distance to coast					-0.398***			-0.339***
					(0.077)			(0.126)
Elevation						-0.018		-0.207
						(0.221)		(0.322)
Ruggedness							-0.057	-0.468***
							(0.044)	(0.067)
Centralized state antiquity (adj.)								-0.088
R^2	0.561	0.061	0.012	0.191	0.038	0.000	0.002	0.707
Observations	627	627	627	627	627	627	627	627

Note: Robust standard errors are in parentheses. All regressions include a constant. Nighttime light intensity growth is the difference between the natural logarithm of nighttime light intensity in 2013 and the natural logarithm of nighttime light intensity in 1992. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. The base climatic zone is Am. Centralized state antiquity (adj.) represents the estimated coefficient of state antiquity adjusted for omitted-variable bias following Oster (2019).

* p<0.1, ** p<0.05, *** p<0.001

Table 5. Robustness: Top-Coding and Lowland Areas

A. Top-coding	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.102***	0.113***	-0.133***	-0.100***	-0.203***	-0.160***
	(0.030)	(0.036)	(0.023)	(0.024)	(0.012)	(0.018)
Ln light intensity 1992					-0.316***	-0.478***
					(0.018)	(0.020)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.018	0.436	0.047	0.584	0.535	0.668
Observations	627	627	627	627	627	627
B. Lowland areas	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.157***	0.121***	-0.062***	-0.063***	-0.155***	-0.122***
	(0.031)	(0.042)	(0.021)	(0.024)	(0.011)	(0.015)
Ln light intensity 1992					-0.412***	-0.511***
					(0.018)	(0.018)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.044	0.382	0.015	0.484	0.681	0.762
Observations	523	523	523	523	523	523

Note: Robust standard errors are in parentheses. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. Control variables include caloric suitability, irrigation suitability, Köppen-Geiger climatic zones, distance to the coast, elevation, ruggedness, and a constant. All columns use nighttime light intensity corrected for top-coding. Panel A includes all districts. Panel B excludes districts whose elevations are above 500 meters.

* p<0.1, ** p<0.05, *** p<0.001

Table 6. Robustness: Latitude

A. As an explanatory variable	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity		-0.212*		-0.200**		-0.089
		(0.119)		(0.087)		(0.064)
Latitude	0.050**	0.277***	-0.126***	0.085	-0.160***	-0.061
	(0.023)	(0.099)	(0.018)	(0.076)	(0.010)	(0.055)
Ln light intensity 1992					-0.326***	-0.475***
					(0.017)	(0.020)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.018	0.436	0.047	0.584	0.535	0.668
Observations	627	627	627	627	627	627
B. As an instrumental variable	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.072**	0.146***	-0.168***	-0.081***	-0.214***	-0.148***
	(0.031)	(0.038)	(0.024)	(0.025)	(0.014)	(0.019)
Ln light intensity 1992					-0.353***	-0.536***
					(0.019)	(0.019)
Control variables	NO	YES	NO	YES	NO	YES
Observations	627	627	627	627	627	627

Note: Robust standard errors are in parentheses. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. Control variables include caloric suitability, irrigation suitability, Köppen-Geiger climatic zones, distance to the coast, elevation, ruggedness, and a constant. Panel A includes latitude as a control variable, Panel B treats latitude as an instrumental variable for centralized state antiquity.
* p<0.1, ** p<0.05, *** p<0.001

Table 7. Conformism

	Gender		Family	
	(1)	(2)	(3)	(4)
Centralized state antiquity	0.009***	0.018*	0.019***	0.036***
	(0.003)	(0.011)	(0.003)	(0.008)
Control variables	NO	YES	NO	YES
R^2	0.006	0.109	0.034	0.069
Observations	1431	1431	1494	1494

Note: Robust standard errors are in parentheses. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. Control variables include caloric suitability, irrigation suitability, Köppen-Geiger climatic zones, distance to the coast, elevation, ruggedness, and a constant. Gender equals 1 if a respondent agrees to the following statement: "*Being a housewife is just as fulfilling as working for pay*". Family equals 1 if a respondent answers "*important*" to the following question: "*How important is the family in your life*?"
* p<0.1, ** p<0.05, *** p<0.001

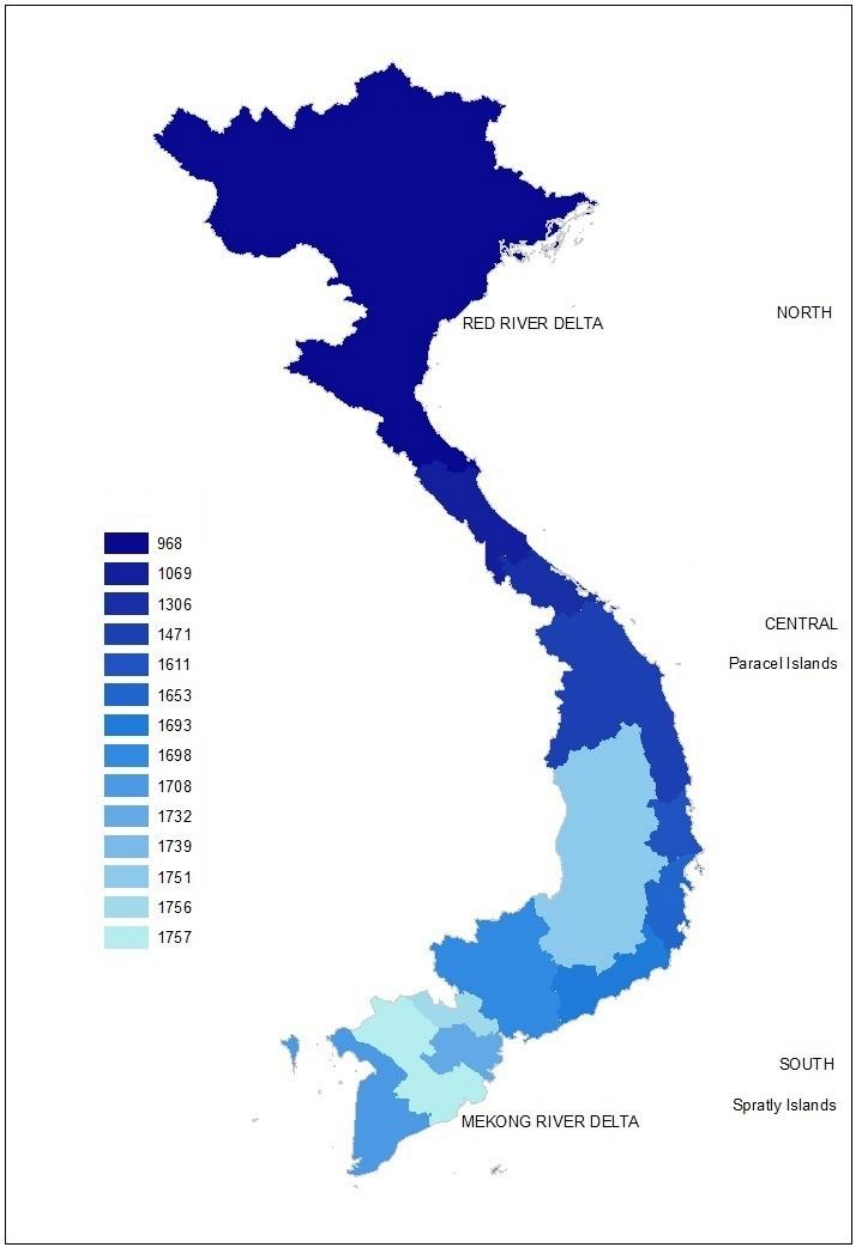
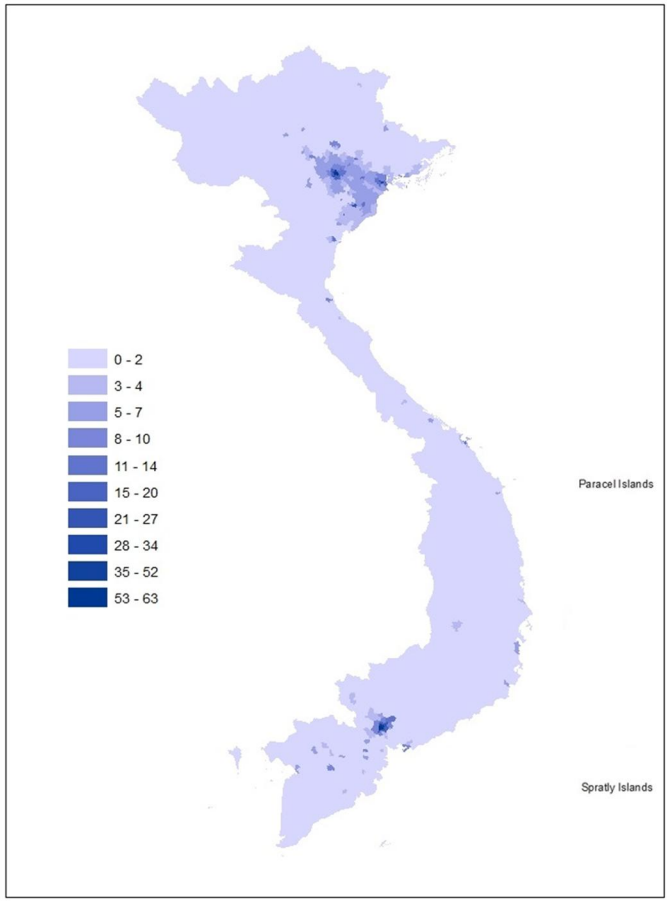
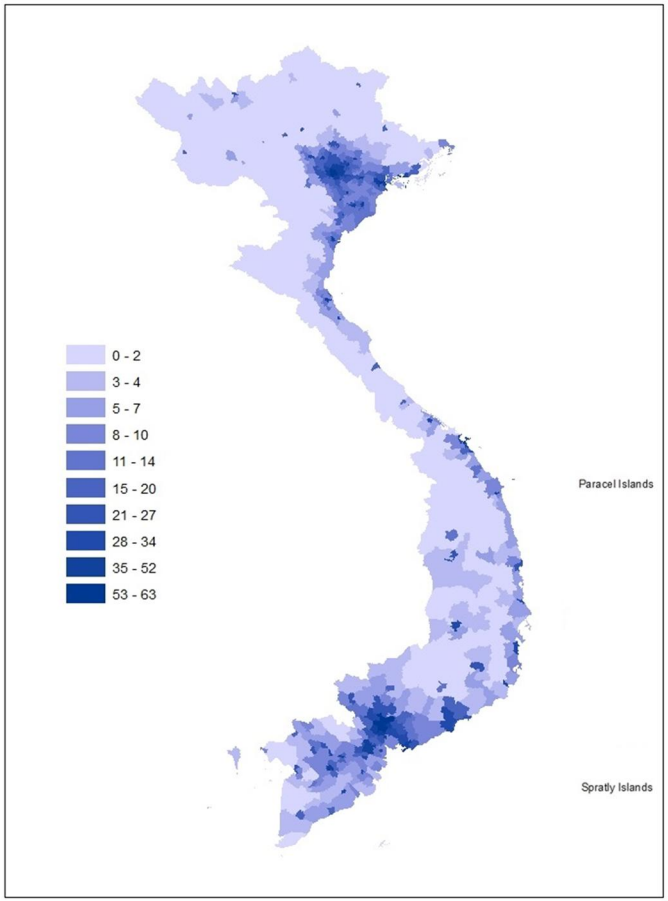


Figure 1. The Vietnamese Southern Advance

Source: Ho, Martinsson, and Olsson (2022)



A. 1992



B. 2013

Figure 2. Nighttime Light Intensity 1992 and 2013

Appendix. Additional tables

Table A1. Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Centralized state antiquity	1						
(2) Caloric suitability	-0.502	1					
	(0.000)						
(3) Irrigation suitability	-0.369	0.151	1				
	(0.000)	(0.000)					
(4) Distance to coast	0.362	-0.592	-0.207	1			
	(0.000)	(0.000)	(0.000)				
(5) Elevation	0.120	-0.204	-0.233	0.685	1		
	(0.003)	(0.000)	(0.000)	(0.000)			
(6) Ruggedness	0.297	-0.346	-0.254	0.645	0.831	1	
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)		
(7) Latitude	0.971	-0.527	-0.404	0.483	0.239	0.381	1
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	

Note: Pearson's correlation coefficients, p-values are in parentheses.

Table A2. Robustness: Chinese Colonization and Spatial Dependence

A. Chinese colonization	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity (adj.)	0.070***	0.095***	-0.048***	-0.033**	-0.094***	-0.078***
	(0.014)	(0.019)	(0.011)	(0.014)	(0.006)	(0.011)
Ln light intensity 1992					-0.316***	-0.478***
					(0.018)	(0.020)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.040	0.452	0.030	0.609	0.572	0.705
Observations	627	627	627	627	627	627
B. Spatial Dependence	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.104**	0.114**	-0.127***	-0.089***	-0.193***	-0.142***
	(0.051)	(0.053)	(0.039)	(0.028)	(0.031)	(0.037)
Ln light intensity 1992					-0.357***	-0.537***
					(0.030)	(0.023)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.272	0.585	0.291	0.713	0.885	0.923
Observations	627	627	627	627	627	627

Note: Panel A: Robust standard errors are in parentheses, centralized state antiquity (adj.) is the numbers of centuries being governed by the Chinese colonization and the centralized state of historical Vietnam. Panel B: Standard errors in parentheses are calculated following Conley (1999) with the assumption that autocorrelation decreases in the distance between district centroids and equals zero for districts that are more than 25 kilometers apart. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. Control variables in both panels include caloric suitability, irrigation suitability, Köppen-Geiger climatic zones, distance to the coast, elevation, ruggedness, and a constant.

* p<0.1, ** p<0.05, *** p<0.001

Table A3. Other mechanisms

A. Western contacts	Light 1992		Light 2013		Growth 1992-2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity		0.097**		-0.171***		-0.216***
		(0.046)		(0.034)		(0.026)
Cochinchina	0.541**	-0.185	1.389***	-0.845***	1.068***	-0.760***
	(0.235)	(0.350)	(0.140)	(0.214)	(0.084)	(0.153)
Ln light intensity 1992					-0.316***	-0.478***
					(0.018)	(0.020)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.008	0.441	0.087	0.623	0.475	0.718
Observations (districts)	627	627	627	627	627	627
B. Public investments	2010		2011		2012	
(million VND per person)	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.122	0.191	0.074	0.107	0.107	0.085
	(0.157)	(0.230)	(0.144)	(0.203)	(0.122)	(0.169)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.011	0.310	0.005	0.251	0.013	0.263
Observations (provinces)	56	56	59	59	59	59
C. State capacity	2011		2012		2013	
(standardized index)	(1)	(2)	(3)	(4)	(5)	(6)
Centralized state antiquity	0.049	0.131	0.109***	0.202***	0.055	0.137**
	(0.034)	(0.084)	(0.031)	(0.056)	(0.034)	(0.064)
Control variables	NO	YES	NO	YES	NO	YES
R^2	0.031	0.182	0.153	0.398	0.039	0.252
Observations (provinces)	63	63	63	63	63	63

Note: Robust standard errors are in parentheses. Centralized state antiquity is the numbers of centuries being governed by the centralized state of historical Vietnam. Control variables in both panels include caloric suitability, irrigation suitability, Köppen-Geiger climatic zones, distance to the coast, elevation, ruggedness, and a constant.
* p<0.1, ** p<0.05, *** p<0.001